Adaptive Explanation Obfuscation via Reinforcement Learning

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***Abstract*—Explainable AI (XAI) methods such as SHAP and LIME tell clients why a model made a decision, but the same explanations help a malicious client extract the model, threatening the owner's intellectual property. We propose a reinforcement learning (RL) defense that sits between the explainer and the client and, for every query, decides how strongly to obfuscate the released explanation. The defender is trained with Proximal Policy Optimization (PPO) in a min-max game against an adversary that builds a surrogate model from the released explanations; its reward is the surrogate's error, weighted by λ, minus the distortion of the explanation, weighted by μ. We evaluate the defense on two pipelines, XGBoost with SHAP on Adult Income and a neural network with LIME on Credit Card Fraud, against an attacker that queries natural data and one that synthesizes queries from a few seed samples, and compare it with four static defenses and the detector PRADA. On Adult Income, the agent raises the adversary's error by 16% over no defense while preserving 97% of the top-3 features. Against synthetic queries on the same dataset, it comes close to PRADA's security at one sixth of its distortion, and unlike PRADA it also defends against attackers that query natural data. On the highly imbalanced Credit Card Fraud data no defense improves security, and the agent learns to leave the explanations nearly intact. The defense adds about 0.34 ms per query.**

***Index Terms*—Explainable AI (XAI), Reinforcement Learning (RL), Model Extraction Attacks, Privacy, Obfuscation.**

I. Introduction

Explainable Artificial Intelligence (XAI) focuses on making complex models and their predictions interpretable, providing insights into the reasoning processes behind machine learning outputs. However, while offering explanations assists legitimate users in comprehending model behavior, it simultaneously creates potential vulnerabilities that adversaries may leverage to replicate models.

A model extraction attack occurs when an adversary attempts to replicate a black box model. If an attack utilizes explanations, it is called an explanation-aware model extraction attack (XaMEA). In such attacks, the adversary sends a massive volume of crafted queries to reveal information about the target model's behavior. The explanations consist of additional information such as feature importance or confidence scores, which the attacker uses to match both the outputs and explanations of a surrogate model to those of the target model. Therefore, when accompanied by explainability, even highly complex models remain vulnerable to extraction attacks.

To address these challenges, we introduce an approach that integrates a reinforcement learning (RL) agent with XAI to dynamically enhance privacy and security without strictly compromising the explanation quality for regular users.

II. Related Work

To contextualize our dynamic obfuscation approach, we review the literature across four key domains: model extraction vulnerabilities, the privacy-explainability trade-off, static defensive mechanisms, and the application of reinforcement learning in adversarial defense.

A. Vulnerabilities in Machine Learning (Model Extraction)

Standard machine learning models exposed via inference APIs are highly susceptible to model extraction (or model stealing) attacks [1], [2]. In these scenarios, adversaries systematically query the prediction API and utilize the resulting input-output pairs to train a functionally equivalent surrogate model [3]. Recent advances have demonstrated that these attacks can achieve both high accuracy and high fidelity representations of the target model [4]. This vulnerability extends beyond simple tabular classifiers, threatening complex sequential recommenders [5] and large language models [6]. Comprehensive surveys of the field emphasize that modern extraction methodologies are becoming increasingly query-efficient, drastically reducing the time and cost required for an adversary to steal proprietary intellectual property [7].

B. The Privacy-Explainability Trade-off and XaMEA

The deployment of XAI methodologies, such as SHAP [8], LIME [9], and Integrated Gradients [10], has become a standard practice for establishing algorithmic transparency. However, the literature increasingly recognizes a strict privacy-explainability trade-off [11]. Explanations inherently leak gradient information and expose local decision boundaries [12]. Milli et al. [13] theoretically and empirically demonstrated that gradient-based explanations allow for rapid, query-efficient model reconstruction.

This inherent vulnerability spawned the development of Explanation-Aware Model Extraction Attacks (XaMEA) [14]. By exploiting feature importances alongside standard outputs, attackers can accelerate the extraction process. Attack frameworks such as AUTOLYCUS [15] leverage model-agnostic

tools like LIME and SHAP to infer decision boundaries with minimal queries, while similar architectures like Megrez [16] effectively target Deep Neural Networks. Furthermore, Shokri et al. [17] established that detailed explanations also exacerbate membership inference risks, confirming that transparency tools uniquely expose models to adversarial exploitation and manipulation [18].

C. Current Defensive Strategies and Their Limitations

Mitigating model extraction traditionally relies on static defense mechanisms. Common techniques include Top-k truncation [5], which limits the informational bandwidth by restricting output dimensions, and prediction generalization, which obscures outputs through deliberate rounding [1]. More proactive defenses attempt to inject controlled, randomized noise into the API predictions [11], [19], often drawing upon differential privacy (DP) frameworks [20] to provide mathematical privacy guarantees for interpretable models [21].

Other state-of-the-art approaches, such as PRADA [22] and adaptive misinformation frameworks [23], [24], analyze incoming query distributions to detect anomalous behavior and intentionally mislead attackers. However, when these static noise or truncation defenses are applied to XAI outputs, they inflict severe utility loss [11]. Static defenses cannot dynamically balance the conflicting needs of maintaining high explanation fidelity for benign users while securing the model against active adversaries.

D. Reinforcement Learning in Adversarial Defense

To effectively combat adaptive adversaries, defensive mechanisms must also be adaptive and unpredictable [25]. Reinforcement Learning (RL) has emerged as a powerful paradigm for addressing complex, dynamic challenges in cybersecurity [26]. Foundational algorithms like Deep Q-Networks (DQN) [27] and Proximal Policy Optimization (PPO) [28] have been successfully adapted for real-time threat detection and network intrusion mitigation [29].

In the specific context of adversarial machine learning, frameworks like Robust Adversarial Reinforcement Learning (RARL) [30] formulate security as a continuous min-max game. This forces the RL agent to learn robust behavioral policies against active perturbations. By extending this game-theoretic RL approach to XAI obfuscation, our system overcomes the core limitations of static privacy mechanisms, allowing the model to dynamically optimize the trade-off between interpretability and security on a per-interaction basis.

III. Proposed System Architecture

The system consists of an inference provider who manages the ML model along with its associated explainability module. The RL agent acts as a protective mechanism that decides how much information to conceal or modify in an explanation.

As illustrated in Fig. 1, the high-level components of the system are:

- AI Model: A model hosted and served by the inference provider. In our evaluations, we utilize both Decision Tree Ensembles (XGBoost) and Deep Neural Networks (DNN) to demonstrate architectural generalizability.
- Input: The query submitted for processing.
- RL agent: Determines how much information to conceal in an explanation to limit model extraction while preserving utility. We utilize a Proximal Policy Optimization (PPO) algorithm for this agent.
- Obfuscation module: Intentionally obfuscates explanations to prevent attackers from learning too much about the model.
- XAI module: Generates baseline explanations for the outputs of the model using frameworks such as SHAP or LIME.

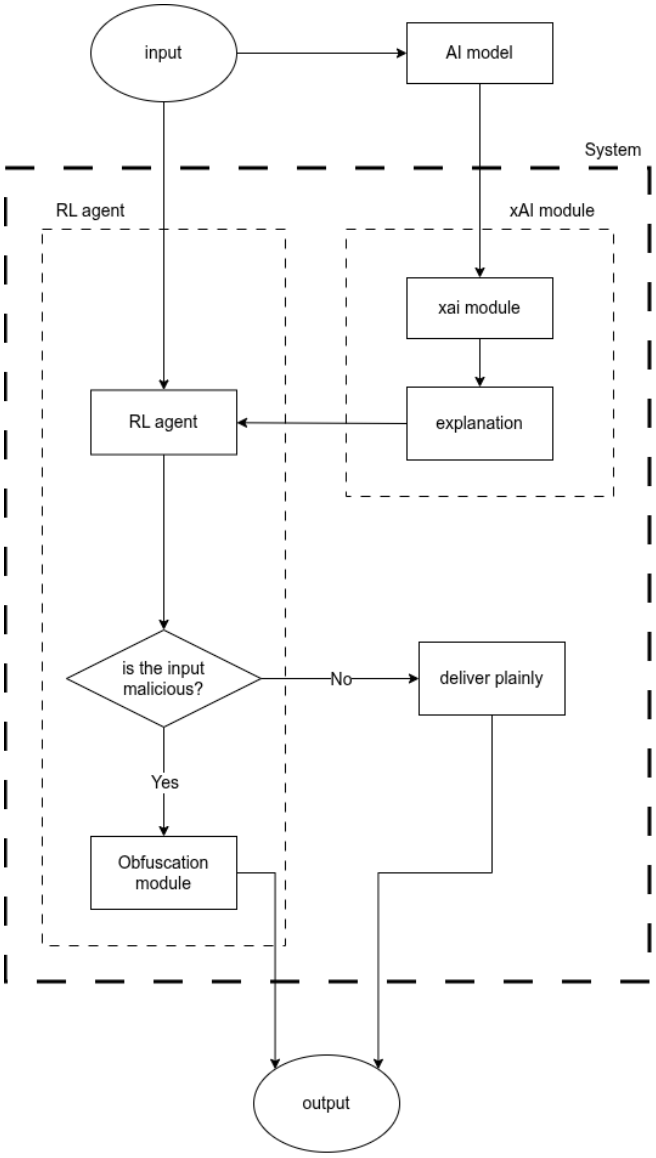


Fig. 1. High-level overview of the system architecture.

The workflow operates as follows (see Fig. 2):

1) An input query is submitted to the system.
2) The input is processed on the model, generating an

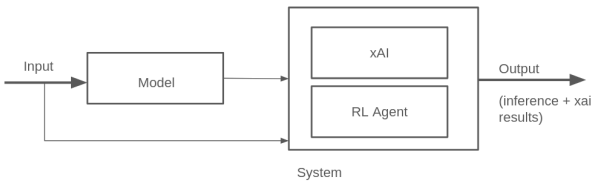


Fig. 2. System workflow: inference model output is explained by the XAI module and selectively obfuscated by the RL agent before returning results.

output.

3) The input is simultaneously sent to the RL agent for analysis.
4) The RL agent determines the degree of obfuscation required from its state, which encodes the current query and, in the rolling-window variant of Eq. (1), the user's preceding queries.
5) The XAI module receives the output and generates a baseline explanation, which is sent to the RL agent.
6) The RL agent decides whether to apply obfuscation to the explanation via the obfuscation module.
7) The processed output and the (potentially obfuscated) explanation are returned to the user.

IV. Game Design for the RL Agent

A. Reinforcement Learning Configuration

Algorithm Selection: To drive the dynamic obfuscation module, we employ the Proximal Policy Optimization (PPO) algorithm [28]. PPO was selected over alternative reinforcement learning algorithms, such as Deep Q-Networks (DQN) or Soft Actor-Critic (SAC), due to its inherent stability and compatibility with continuous action spaces. Because our obfuscation intensity a_t is a continuous variable defined in the range $[0, 1]$, value-based methods like DQN, which require discrete action spaces, are unsuitable without arbitrary discretization. While SAC accommodates continuous actions, PPO's clipped surrogate objective function prevents destructively large policy updates, ensuring stable and monotonic improvement during the delicate min-max adversarial training process.

State-Space Formulation and Network Architecture: The RL agent must make contextual decisions based on the current query and the history of adversarial interactions. The user's query history grows with every query, whereas an MLP policy requires an input of fixed size. We therefore encode the history in fixed size as a *rolling window* of the user's last k queries. Each query is first standardised with the training-set statistics, $z_t = (x_t - \bar{x})/\sigma_x$, and the state is the concatenation of the k most recent standardised queries and the normalized step counter:

$$s_t = \left[z_{t-k+1} \parallel \cdots \parallel z_{t-1} \parallel z_t \parallel \frac{t}{T_{max}} \right] \tag{1}$$

Where t is the current query step and T_{max} represents the maximum allowed queries in a session. At the start of a session ($t < k$), missing slots are filled with zeros (the feature mean). The size of s_t, $kd + 1$ for d features, is independent of t. The window preserves the order and recency of the user's queries, so the policy can condition on local query patterns such as bursts of near-duplicate or slightly perturbed queries, which session-wide averages would smooth out. We use $k = 8$. This combined vector is processed by the PPO agent, which is parameterized using a standard Multi-Layer Perceptron (MLP) architecture. The MLP network acts as both the actor, outputting the continuous action a_t, and the critic, estimating the state-value function to guide policy updates.

To measure what the history contributes, we also train agents with *no history*. Their state contains only the current query and the step counter:

$$s_t = \left[z_t \parallel \frac{t}{T_{max}} \right] \tag{2}$$

A no-history agent therefore decides the obfuscation of each query from that query alone; it does not see any of the user's earlier queries. This is the state of our original formulation. Throughout the paper, "no history" refers to agents with the state of Eq. (2), and "rolling window" to agents with the state of Eq. (1); everything else, including the reward and the training procedure, is identical.

Convergence Behavior: Each training episode is one simulated session of $T_{max} = 1000$ queries, drawn in random order from a pool of 5,000 training samples; a new random session is drawn for every episode, so the agent cannot memorise a fixed query sequence. Operating over 50,000 environmental timesteps (50 episodes) with a learning rate of 3×10^{-4}, the agent learns a defensive policy. Empirical learning curves derived from the environment's rolling episode rewards indicate that the agent steadily increases its reward, confirming that PPO effectively managed the volatile signals generated by the adversary's simultaneously evolving surrogate model. However, 50,000 timesteps amount to only 50 sessions, and on Credit Card Fraud the agents had not converged by then: for $\lambda = 1$, $\mu = 0.05$ and the no-history state, the average episode reward was still rising steeply, from about 23 in the first ten sessions to about 37 in the last ten. We therefore retrain part of the grid for 500,000 timesteps (500 sessions; Section V-A). With this budget the reward of the no-history agent levels off after about 200 sessions, at about 46; the rolling-window agent, whose state is eight times larger, is still improving slowly after 500 sessions. The longer training changes the Credit Card Fraud results only in the distortion, not in the security (Section V-C).

The RL agent is trained using a game-based framework designed to enhance model security while maintaining explanation utility. The setup simulates a continuous adversarial scenario, as depicted in Fig. 3.

B. Min-Max Game Formulation

Step 1: State at time t

After t queries, the defender has received the user's queries $x_1, \ldots, x_t$ and the model's outputs $y_i = M(x_i)$ for them. Since this history grows with every query, the agent observes it through the fixed-size state s_t: the rolling window of Eq. (1)

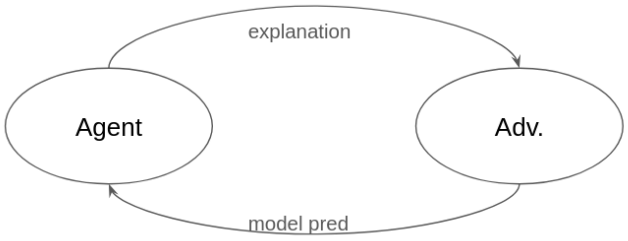


Fig. 3. Game-based training loop: the RL agent obfuscates explanations; the adversary trains a surrogate model, and rewards/penalties are assigned based on security and explanation utility.

or, for the no-history agents, Eq. (2). The outputs are a deterministic function of the queries, so both encodings keep the queries only.

Step 2: RL Agent Action

The RL agent determines an obfuscation function f_{obfus} to apply to the true explanation E_{true}. The action a_t is a continuous variable $a_t \in [0, 1]$ representing the intensity of obfuscation. The applied transformation is defined as:

$$E_{output} = (1 - a_t)E_{true} + a_t \cdot \epsilon \tag{3}$$

Where ϵ represents Gaussian noise scaled by the standard deviation of the true explanations ($\epsilon \sim \mathcal{N}(0, \sigma_E^2)$). If $a_t = 0$, $E_{output} = E_{true}$ (full explanation). If $a_t = 1$, E_{output} relies entirely on the generated noise (full obfuscation).

Step 3: Adversary Action

The adversary leverages E_{output} to construct a surrogate model S targeting the original model M. The adversary trains S to minimize its error $\mathcal{E}_t$, the adversary error of Section V-B:

$$\min \mathcal{E}_t(S(x), M(x)|E_{output}) \tag{4}$$

The surrogate is trained incrementally: it is re-initialised at the start of every episode and updated with one gradient step per query on the input $[x_t, E_{output}]$ and label $M(x_t)$. $\mathcal{E}_t$ is the surrogate's cross-entropy with the target's labels on its $W = 32$ most recent queries, measured before the update on x_t.

Step 4: Reward Function

The reward function R_t balances security (maximizing the adversary error) and utility (minimizing the explanation distortion).

$$R_t = \lambda \cdot \mathcal{E}_t(S, M) - \mu \cdot \frac{||E_{true} - E_{output}||}{||E_{true}||} \tag{5}$$

Where λ and μ are non-negative scalar hyperparameters weighing the importance of model security and explanation fidelity, respectively. The distortion term is normalized by $||E_{true}||$, as in the evaluation metric of Section V-B, so that it is on the same scale as $\mathcal{E}_t$.

V. Experimental Evaluation

A. Experimental Setup

To evaluate the proposed methodology and demonstrate its structural generalizability, we implemented two distinct machine learning pipelines:

1) **Adult Income (XGBoost + SHAP):** A black-box XGBoost target model providing interpretability via SHAP TreeExplainer.
2) **Credit Card Fraud (DNN + LIME):** A Deep Neural Network target model providing interpretability via LIME (Local Interpretable Model-agnostic Explanations).

In both environments, the adversary was designed as an adaptive attacker aiming to replicate the target model using a surrogate Multi-Layer Perceptron (MLP) Classifier configured with two hidden layers of 64 and 32 neurons. The RL defending agent was trained independently for each pipeline using the PPO algorithm over 50,000 timesteps.

Attackers. The *natural-query* attacker queries natural data in random order, as during training. The *synthetic-query* attacker owns only 10 seed samples and generates new queries by perturbing earlier ones with random-sign steps of $0.1\sigma_x$ on a random subset of features, in the spirit of the augmentation-based attacks that PRADA [22] is designed to detect. All strategies are evaluated on the same three sessions of 500 held-out queries per attacker.

Agents. Following the formulation of Section IV-B, we vary both weights: $\lambda \in \{0, 0.1, 0.25, 0.5, 0.75, 1, 1.5, 2, 3, 5\}$ at each $\mu \in \{0, 0.01, 0.05, 0.1, 0.2, 0.5\}$ (except $\lambda = \mu = 0$). Each agent is trained with the rolling-window state of Eq. (1) and, as an ablation, with the no-history state of Eq. (2), over three random seeds, for 50,000 timesteps; reported values are averages over the seeds. Because the agents on Credit Card Fraud had not converged after 50,000 timesteps (Section IV-A), we additionally train the agents with $\mu = 0.05$ and $\lambda \in \{0, 0.5, 1, 2, 5\}$, for both states and three seeds (60 agents), for 500,000 timesteps, on both datasets.

Implementation and hardware. The defense is implemented in Python 3.12 with Gymnasium and the PPO implementation of Stable-Baselines3 (PyTorch). The target models are built with XGBoost and scikit-learn, the explanations with SHAP and LIME, and the adversary's surrogate with scikit-learn. The explanations of all training and evaluation queries are computed once before training, so no SHAP or LIME call is made while the agents are trained. All experiments ran on one CPU node of a high-performance computing cluster, with two 32-core AMD EPYC 7543 processors (64 cores in total) and 1 TB of memory; no GPU was used. Each agent is trained and evaluated in its own single-threaded process, 64 at a time. Training one agent takes about 4 minutes for 50,000 timesteps and about 36 minutes for 500,000 timesteps; the full grid of 708 agents, including the evaluation of all agents and baselines, completed in 26 minutes.

B. Evaluation Metrics

We benchmark the defensive policies across three primary metrics to assess the balance between adversarial security and user interpretability:

1. Average Adversary Error ($\mathcal{E}_t$): Measures the cross-entropy loss of the adversary's surrogate model. Higher values indicate successful defense against extraction. (We rename this metric from "extraction loss" because the defender maximizes it, and denote it by $\mathcal{E}_t$ instead of $L_{extract}$.)

2. Average Explanation Distortion (L2 Distortion): The normalized L2 norm between the true explanation and the obfuscated output, representing the raw geometric distortion.

3. Semantic Interpretability Metrics: To ensure the obfuscation policy does not destroy the inherent meaning of the explanations, we evaluate two semantic utility metrics:

- *Spearman's Rank Correlation (ρ):* Measures the monotonic relationship between the feature importance rankings of the original explanation E_{true} and the obfuscated explanation E_{output}. A higher ρ indicates that the relative hierarchy of feature importance is preserved.
- *Top-k Feature Agreement:* Calculates the intersection of the k most critical features before and after obfuscation, defined as $\frac{|S_{true} \cap S_{output}|}{k}$, where S represents the set of indices for the top k features. We report results for $k = 3$.

C. Results and Baselines

The trained PPO agents were benchmarked against multiple static defense strategies prevalent in the literature: Top-k truncation ($k = 3$), Gaussian Noise addition (level=0.5), Precision Reduction (2 decimals), and Random Subset omission ($p = 0.5$). We also compare against the state-of-the-art detector PRADA [22], implemented from its published algorithm on standardised features. As PRADA detects rather than obfuscates, we use it as a gate: explanations are exact until the client is flagged and fully obfuscated ($a_t = 1$) afterwards. Its threshold δ is the largest value that raises no alarm on benign sessions.

Correction. In the original submission, the adversary error was set to a constant 1.0 whenever the adversary's 32-query window contained a single class, which is almost always the case on the highly imbalanced Credit Card Fraud data; this produced its uniform values of 0.98–0.99. All results in this version are recomputed with the corrected metric.

As demonstrated in Table I, static methods struggle to balance the tradeoff effectively on the SHAP pipeline. While Gaussian Noise and Random Subset methods exhibit higher security, they inflict severe utility loss (1.0688 and 0.5883, respectively), rendering the explanations functionally useless for benign users. In contrast, the RL Agent ($\mu = 0.05$) successfully increases security (0.2717) over the baseline (0.2345, +16%) while maintaining an exceptional explanation fidelity (Top-3 Agreement ≈ 0.97). It thus nearly matches the security of Random Subset (0.2758) at one seventh of its distortion, and is more secure than Gaussian Noise at less than a tenth of its distortion. Fig. 4 shows the complete trade-off for both datasets.

Varying λ. Fig. 5 reports all four metrics as a function of the security weight λ. On Adult Income, increasing λ raises the adversary error from the undefended 0.2345 to about 0.28 while distortion grows and the rank metrics decrease, i.e. λ moves the agent along the security–utility trade-off. Since multiplying R_t by a positive constant does not change the optimal policy, the trade-off depends on the ratio λ/μ, which our grid largely confirms. For example, the no-history agents with $(\lambda, \mu) = (1, 0.05)$ and $(2, 0.1)$, both with $\lambda/\mu = 20$, reach adversary errors of 0.272 and 0.270 at distortions of 0.08 and 0.09, and the five agents with $\lambda/\mu = 10$, from $(0.1, 0.01)$ to $(5, 0.5)$, all reach adversary errors between 0.244 and 0.257 at distortions between 0.03 and 0.06. The exception is the largest weight, $\lambda = 5$: there PPO reaches a worse trade-off than at a smaller λ with the same ratio, e.g. 0.259 at a distortion of 0.34 for $(5, 0.1)$, against 0.281 at 0.17 for $(0.5, 0.01)$. PPO is not exactly invariant to the scale of the reward, because its value function must fit returns that grow with λ and μ.

Comparison with PRADA. PRADA never flags the natural-query attacker, whose queries are, by construction, indistinguishable from those of a benign client (Tables I and II). Against the synthetic-query attacker it is effective but coarse (Table III, which lists no defense, the most secure static baseline, Random Subset, and PRADA, together with the RL agents): it flags the client at query 101–102, the earliest point its statistical test allows, and fully obfuscates everything afterwards. On Adult Income, the RL agent ($\lambda = 1$, $\mu = 0.01$, no history) comes close to PRADA's security (0.1898 vs. 0.2101) at one sixth of its distortion (0.24 vs. 1.49), while preserving the feature ranking ($\rho = 0.85$ vs. 0.20; ρ is not shown in Table III). The two defenses are complementary. Unlike detection-based defenses such as PRADA, which cannot identify attackers that query natural data, the RL agent protects against both attacker types. Against synthetic-query attackers on Adult Income, PRADA attains the highest security but destroys the explanations once triggered, whereas the RL agent attains comparable security at a fraction of the distortion. On Credit Card Fraud, PRADA flags the attacker as well, but its obfuscation brings no security gain (0.0623 vs. 0.0627 without defense) at a distortion of 1.44; as with the natural-query attacker, no defense helps on this dataset, for the reason discussed below. PRADA, in turn, releases exact explanations to clients it does not flag, whereas the RL agent studied here obfuscates for every client (see below).

Effect of the query history. Across both datasets and attackers, the rolling-window agent does not reach a better trade-off than the no-history agent of Eq. (2): at equal distortion, the no-history agent is at least as secure (Fig. 4and Tables I–III). This is expected in the game studied here: every training session is an extraction attempt with independently drawn queries, so earlier queries carry no information about the current one, while the larger input makes the policy harder to learn. Since the rolling-window state is eight times larger, it might need more training than the 50,000 timesteps. The 500,000-timestep agents, trained because the Credit Card

TABLE I
SECURITY VS. DISTORTION TRADE-OFF: ADULT INCOME DATASET (XGBOOST + SHAP), NATURAL-QUERY ATTACKER

Strategy	Adversary Error ↑	Expl. Distortion ↓	Spearman's ρ ↑	Top-3 Agreement ↑
Static Baselines				
No Defense	0.2345	0.0000	1.0000	1.0000
Precision Reduction (2 dec)	0.2343	0.0062	0.9982	0.9949
Top-k ($k = 3$)	0.2438	0.1474	0.9411	1.0000
Gaussian Noise (lvl=0.5)	0.2521	1.0688	0.5901	0.7211
Random Subset (p=0.5)	0.2758	0.5883	0.5575	0.5471
PRADA [22]	0.2345	0.0000	1.0000	1.0000
RL Adaptive Agents ($\lambda = 1.0$), rolling-window state, Eq. (1)				
RL ($\mu = 0.1$)	0.2536	0.1607	0.9128	0.9447
RL ($\mu = 0.05$)	0.2651	0.3763	0.8140	0.8977
RL ($\mu = 0.01$)	0.2795	0.6804	0.6876	0.8396
RL ($\mu = 0.0$)	0.2826	0.7535	0.6588	0.8264
RL Adaptive Agents ($\lambda = 1.0$), no history, Eq. (2)				
RL ($\mu = 0.1$)	0.2552	0.0529	0.9613	0.9793
RL ($\mu = 0.05$)	**0.2717**	**0.0813**	**0.9351**	**0.9704**
RL ($\mu = 0.01$)	0.2912	0.4182	0.7567	0.8613
RL ($\mu = 0.0$)	0.3007	0.9864	0.5178	0.7407

TABLE II
SECURITY VS. DISTORTION TRADE-OFF: CREDIT CARD FRAUD DATASET (DNN + LIME), NATURAL-QUERY ATTACKER

Strategy	Adversary Error ↑	Expl. Distortion ↓	Spearman's ρ ↑	Top-3 Agreement ↑
Static Baselines				
No Defense	0.0866	0.0000	1.0000	1.0000
Precision Reduction (2 dec)	0.0851	0.5643	0.7728	0.4438
Top-k ($k = 3$)	0.0822	0.7632	0.5287	1.0000
Gaussian Noise (lvl=0.5)	0.0812	0.7064	0.7967	0.5256
Random Subset (p=0.5)	0.0933	0.6977	0.6469	0.4987
PRADA [22]	0.0866	0.0000	1.0000	1.0000
RL Adaptive Agents ($\lambda = 1.0$), rolling-window state, Eq. (1)				
RL ($\mu = 0.1$)	0.0863	0.3000	0.8524	0.8082
RL ($\mu = 0.05$)	0.0859	0.4240	0.7856	0.7401
RL ($\mu = 0.01$)	0.0848	0.5319	0.7295	0.6859
RL ($\mu = 0.0$)	0.0847	0.5440	0.7231	0.6788
RL Adaptive Agents ($\lambda = 1.0$), no history, Eq. (2)				
RL ($\mu = 0.1$)	0.0864	0.0250	0.9913	0.9840
RL ($\mu = 0.05$)	0.0864	0.0674	0.9739	0.9550
RL ($\mu = 0.01$)	0.0869	0.2622	0.8806	0.8186
RL ($\mu = 0.0$)	0.0870	0.3842	0.8186	0.7484

TABLE III
SYNTHETIC-QUERY ATTACKER

Strategy	Adult Income	Credit Card
	Adversary Error ↑ / Distortion ↓	
No Defense	0.1259 / 0.00	0.0627 / 0.00
Random Subset (p=0.5)	0.1808 / 0.59	0.0685 / 0.70
PRADA [22]	0.2101 / 1.49	0.0623 / 1.44
RL ($\lambda = 1$), rolling-window state, Eq. (1)		
RL ($\mu = 0.05$)	0.1530 / 0.24	0.0633 / 0.30
RL ($\mu = 0.01$)	0.1742 / 0.55	0.0638 / 0.38
RL ($\lambda = 1$), no history, Eq. (2)		
RL ($\mu = 0.05$)	0.1567 / 0.08	0.0629 / 0.12
RL ($\mu = 0.01$)	0.1898 / 0.24	0.0634 / 0.37

Fraud agents had not converged (Section IV-A), also test this, and the gap does not close; it widens. At $\lambda = 1$, $\mu = 0.05$ on Adult Income, ten times more training raises the adversary error of the no-history agent from 0.2717 to 0.3015 (+29% over no defense) at a distortion of 0.14 ($\rho = 0.88$, Top-3 agreement 0.95), but that of the rolling-window agent only from 0.2651 to 0.2671, at a distortion of 0.33. Against the synthetic-query attacker, the no-history agent reaches 0.1991 at a distortion of 0.19, and the rolling-window agent 0.1582 at 0.27. The rolling window is therefore not held back by the training budget. The history encoding becomes relevant when benign and malicious clients are served by the same agent, so that past queries indicate which kind of client is being served; we leave this setting to future work. Relatedly, as the agent is trained only on extraction sessions, it applies the same obfuscation to benign clients, who therefore bear the distortion reported above; training on mixed sessions, rewarding exact explanations for benign clients, is the natural remedy.

The evaluation on the Credit Card Fraud dataset (Table II) further demonstrates the severe limitations of static defenses when applied to LIME explanations. While Top-k Truncation ($k = 3$) maintains perfect Top-3 Agreement, it completely disrupts the overall feature ranking ($\rho \approx 0.53$). Conversely, Precision Reduction preserves general ranking ($\rho \approx 0.77$) but loses over half of the most critical top-3 features (Agreement ≈ 0.44). Our adaptive policy ($\lambda = 1.0, \mu = 0.05$, no history) maintains high rank correlation ($\rho \approx 0.97$) and preserves the most critical features (Top-3 Agreement ≈ 0.96). However, on this dataset no defense, static or adaptive, yields a meaningful security gain: the adversary error stays between 0.081 and 0.093 for every strategy. Fraud accounts for only 0.17% of the transactions, so sessions of natural queries contain almost no positive cases and a surrogate that predicts the majority class imitates the target well regardless of the explanations. The RL agents learn that obfuscation does not pay here and keep the distortion low. Training until the reward converges (500,000 timesteps, $\lambda = 1$, $\mu = 0.05$) confirms this: the adversary error stays the same (0.0866 vs. 0.0864 at 50,000 timesteps for the no-history agent, 0.0867 vs. 0.0859 for the rolling-window agent), while the distortion falls from 0.067 to 0.039 and from 0.424 to 0.264, respectively. The converged agents obfuscate less, because obfuscating more buys no security. Defending explanations of models trained on highly imbalanced data therefore requires a threat model with class-balanced queries, which we leave to future work.

VI. Discussion and Practical Deployment

A. Addressing Computational Overhead

A primary concern when integrating adaptive defense mechanisms into production machine learning pipelines is the introduction of unacceptable latency during real-time inference. Traditional reinforcement learning training is computationally intensive; however, the deployment phase of our proposed architecture introduces little overhead.

The trained PPO actor is parameterized as a lightweight Multi-Layer Perceptron (MLP). During inference, calculating the obfuscation intensity a_t requires only a single forward pass through this shallow neural network. Table IV lists the time per query for each defense. Without a defense, a query takes 0.91 ms on Adult Income (0.30 ms prediction, 0.61 ms TreeSHAP explanation) and 15.29 ms on Credit Card Fraud (0.20 ms prediction, 15.09 ms LIME explanation). The static baselines are listed together, as each adds at most 0.02 ms. All times are medians, measured with one thread per process on the node described in Section V-A. Prediction and explanation were timed once on 100 queries, served one at a time, while the node was otherwise idle; the defense's time was measured on every evaluation query, while other single-threaded evaluation jobs shared the node. The absolute times are therefore conservative and depend on the hardware, whereas the percentages, which compare two measurements on the same machine, carry over to other deployments. The RL defense (state encoding, policy forward pass and obfuscation) takes 0.34 ms, less than PRADA. On Credit Card Fraud, where the LIME explanation dominates, it makes a query only 2.2% slower; with the much faster TreeSHAP it makes a query 37% slower, yet a query is still served in about 1.2 ms. Therefore, the dynamic obfuscation layer can be deployed as an API gateway proxy without violating strict latency service-level agreements (SLAs).

TABLE IV
Time per Query (Median, One CPU Core); Increase Relative to No Defense

	Adult (SHAP)			Credit (LIME)		
Defense	**Added (ms)**	**Total (ms)**	**Increase (%)**	**Added (ms)**	**Total (ms)**	**Increase (%)**
No defense	–	0.91	–	–	15.29	–
Static baselines	≤0.02	≤0.93	≤2.5	≤0.02	≤15.31	≤0.2
PRADA [22]	0.44	1.35	48.6	0.46	15.75	3.0
RL, no history	0.34	1.24	37.3	0.34	15.63	2.2
RL, rolling window	0.34	1.24	37.4	0.35	15.63	2.3

B. Dynamic Threat Posturing

The ablation study demonstrates that a single RL architecture can accommodate diverse security postures. In a practical deployment, a centralized security orchestration system could dynamically swap the active RL agent weights based on the current global threat level. During periods of normal operation, the system can utilize a high-utility agent ($\mu = 0.05$). If an intrusion detection system (IDS) flags an ongoing distributed XaMEA attack, the gateway can instantly pivot to a high-security agent ($\mu = 0.01$), actively stifling the extraction attempt while still providing baseline, albeit noisy, interpretability.

VII. Advantages of the Proposed Solution

The dynamic integration of the RL agent provides several technical benefits:

- **Enhanced Security:** Increases the error of explanation-aware extraction attacks; on Adult Income, the adversary error rises by 16% after 50,000 training timesteps and by 29% after 500,000.
- **Adaptive Explanations:** Chooses the obfuscation level for every query instead of applying a fixed transformation. The state can also encode the user's preceding queries, although in the setting studied here this did not improve the trade-off.
- **Balanced Utility:** At this security level the explanations remain faithful (Top-3 agreement 0.97, $\rho = 0.94$), whereas Random Subset, the static defense with comparable security, distorts them seven times more.
- **Robustness:** The RL agent adapts over time and applies unchanged across varied architectures (Tree Ensembles and DNNs) and XAI methods (SHAP and LIME); where obfuscation cannot improve security, as on the highly imbalanced Credit Card Fraud data, it learns to preserve explanation utility instead.

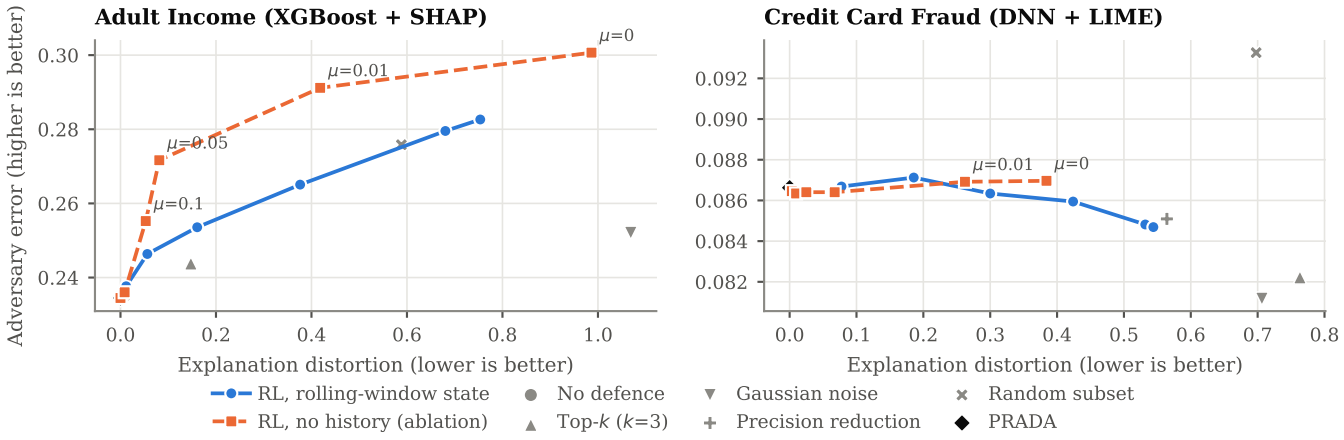


Fig. 4. Security versus Distortion trade-off for both datasets (natural-query attacker). RL agents use $\lambda = 1$ with μ varied, averaged over three seeds; static baselines and PRADA are shown for reference.

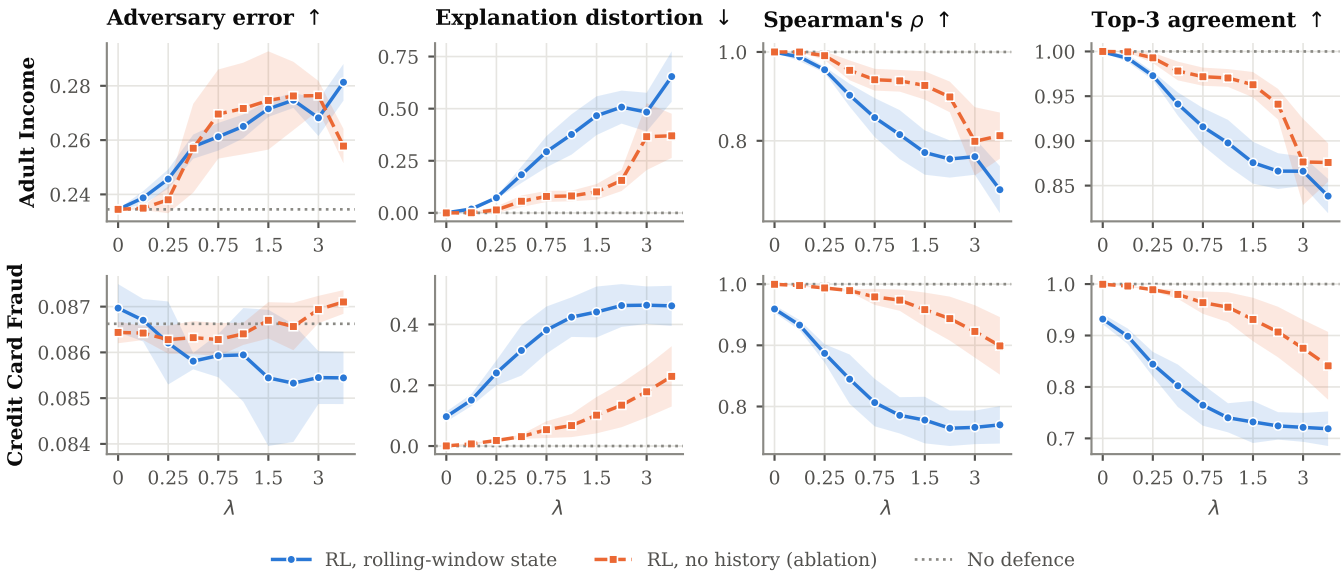


Fig. 5. All four metrics as a function of the security weight λ ($\mu = 0.05$, natural-query attacker), for the rolling-window state(Eq. (1)) and the no-history state (Eq. (2)); mean $\pm$ standard deviation over three seeds. Note the narrow adversary-error range on Credit Card Fraud.

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